Power Converter Analysis and Design

ASSIGNMENT - 5

1. Consider only one inverter leg as shown in Figure 1, where the output current lags $(v_{Ao})_1$ by an angle ϕ, as shown in Figure 2a. and o is the fictitious midpoint of the DC input. Because of the blanking time t_Δ, the instantaneous error voltage v_ϵ, is plotted in Figure 2b, where,

$$v_\epsilon = (v_{Ao})_{ideal} \ - \ (v_{Ao})_{actual}$$

Each v_ϵ pulse, either positive or negative, has an amplitude of V_d and a duration of t_Δ. In order to calculate the low-order harmonics of the fundamental frequency in the output voltage due to blanking time, these pulses can be replaced by an equivalent rectangular pulse (shown dashed in Figure 2b) of amplitude K whose volt-second area per half-cycle equals that of v_ϵ pulses.

Derive the following expression for the harmonics of the fundamental frequency in v_{Ao}, introduced by the blanking time:

$$(\hat{V_{Ao}})_h = \frac{4}{\pi h} V_d t_\Delta f_s \quad (h = 1, 3, 5, ...)$$

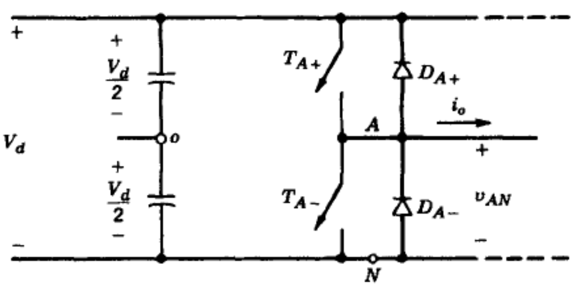

Figure 1: One-leg switch-mode inverter [R1]

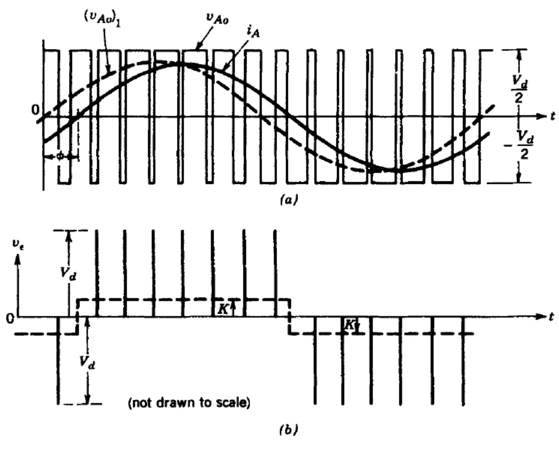

Figure 2: Dead-time effect [R1]

2. Dead-time in an inverter causes

   (a) loss of fundamental voltage at the output of the inverter.

   (b) reduction in switching loss of the inverter.

   (c) increase in switching frequency, effectively, of the power devices.

   (d) generation of low-order harmonics of fundamental, at the output.

3. (a) A three-leg, two-level, three-phase voltage source inverter (VSI) employs sinusoidal pulse-width modulation (SPWM) with the modulation functions defined as:

$$
\begin{aligned}
d_1 &= 1 - d_4 = \frac{1}{2} + \frac{m}{2}\sin(\omega t), \\
d_3 &= 1 - d_6 = \frac{1}{2} + \frac{m}{2}\sin\left(\omega t - \frac{2\pi}{3}\right), \\
d_5 &= 1 - d_2 = \frac{1}{2} + \frac{m}{2}\sin\left(\omega t + \frac{2\pi}{3}\right),
\end{aligned} \tag{1}
$$

   where d_k denotes the instantaneous duty ratio of the k^{th} switch, m is the modulation index, and T_s is the switching period. Each switch incorporates a rising edge delay of 4% of the switching period to prevent through-conduction. Determine the maximum achievable modulation index, such that the inverter operates within the linear modulation range. Determine the maximum peak fundamental line-to-line voltage that can be achieved under linear modulation range.

   (b) For the same inverter configuration, the modulation scheme is modified as shown in (2), incorporating a third-harmonic injection component while maintaining the same dead-time duration of 4% of T_s. Determine the maximum achievable modulation index, such that the inverter operates within the linear modulation range. Determine the maximum peak fundamental line-to-line voltage that can be achieved under linear modulation range.

$$
\begin{aligned}
d_1 &= 1 - d_4 = \frac{1}{2} + \frac{m}{2}\sin(\omega t) + \frac{m}{12}\sin(3\omega t), \\
d_3 &= 1 - d_6 = \frac{1}{2} + \frac{m}{2}\sin\left(\omega t - \frac{2\pi}{3}\right) + \frac{m}{12}\sin(3\omega t), \\
d_5 &= 1 - d_2 = \frac{1}{2} + \frac{m}{2}\sin\left(\omega t + \frac{2\pi}{3}\right) + \frac{m}{12}\sin(3\omega t).
\end{aligned} \tag{2}
$$

4. A three-phase, full-bridge voltage source inverter (VSI) delivers power to a balanced, Y-connected three-phase grid operating at a line-to-line RMS voltage of 415 V and a frequency of 50 Hz. The inverter is supplied from a 1000 V DC link and is interfaced with the grid through a per-phase series inductance of 1 mH. The inverter employs unipolar sinusoidal pulse-width modulation (SPWM) with a carrier frequency ($f_s = 1 \div T_s$) of 10 kHz. The grid follows an RYB phase sequence, with the R-phase voltage defined as

$$v_r(t) = \hat{V}_r \sin(\omega_o t)\,\text{V}.$$

   The inverter operates with an output power of 10 kW, delivering current to the grid at a lagging power factor of 0.9. A dead-time interval corresponding to 4 % of the switching period ($t_\Delta = 0.04T_s$) is incorporated in the modulation process to account for device commutation overlap.

i) Determine the RMS value of the fundamental component of the current injected per phase into the grid.

ii) Compute the RMS value of the fundamental voltage drop across the interfacing inductor.

iii) Determine the RMS value of the fundamental per-phase voltage generated by the inverter.

iv) Derive an analytical expression for the R-phase modulating signal $m_r(t)$, accounting for the specified dead-time, assuming a carrier waveform of unit amplitude.

5. A three phase cascaded H-bridge converter with 5 cells in each phase supplies 5 MW power and 5 MVAR reactive power to a load. The DC link voltage of each cell is 2 kV.

   i) How many levels will be presented in the pole voltage waveform of the converter? ________.

   ii) What will be the peak fundamental pole voltage of the converter if sinusoidal pulse width modulation is used with 0.9 modulation index? ________.

   iii) What will be the device voltage rating? ________.

   iv) The root-mean-square value of converter output current is ________.

   v) What will be the device current rating? ________.

6. The main disadvantage of using level-shift PWM in a cascaded H-bridge converter is

   a) Unequal loss distribution

   b) Higher device voltage rating

   c) Higher device current rating

   d) High $\frac{dv}{dt}$ stress

7. Consider a cascaded H-bridge converter with 10 number of cells in each phase of the converter and having a DC-link voltage of each cell as V_{dc}. The converter can produce a maximum balanced line voltage peak as,

   a) $5V_{dc}$ b) $10V_{dc}$ c) $20V_{dc}$ d) $40V_{dc}$

8. It is desired to construct a cascaded H-bridge (CHB) multilevel inverter for a three-phase induction motor rated at 3 kV, 250 kW. It is decided to use five cells.

   (a) What is the minimum DC voltage required per cell so as to enable operation only in the linear modulation range? (level-shifted carriers can be considered)

   (b) How many voltage levels will therefore be available?

   (c) Suppose that we decide to implement staircase modulation for this system with a DC bus voltage per cell of 500 V. For a voltage waveform arrived by distributing the angles to be chosen between 0° and 90° uniformly between themselves and the extremities (excluding both extremities as candidates for switching instants), find:

i. The fundamental amplitude.

ii. The total harmonic distortion (THD) of the voltage waveform, considering up to the first five dominant harmonics only.

9. In a cascaded H-bridge (CHB) multilevel inverter structure used to supply a three-phase load, three cells are proposed to be used for each phase.

   (a) How many voltage levels can be obtained?

   (b) How many power devices (diodes as well as controllable switches) are used in total for all three phases?

   (c) How many carrier waveforms would be needed for a level-shifted PWM scheme? If phase-shifted carriers are used, how many are needed?

   (d) If space vector modulation (SVM) is considered, how many zero states are available?

10. Consider a multilevel inverter constructed using H-bridge cells, with one cell employed per phase. The DC-link voltages of all cells are equal to E. Determine the locations of the space vectors corresponding to the switching states $(0, -E, -E)$ and $(E, 0, 0)$ in the space vector diagram of this system.

References

[R1] N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics: Converters, Applications, and Design*, 3rd ed. Hoboken, NJ, USA: Wiley, 2003.